CHETAN RAO SONOO

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Education

Sunway University Sept 2023 – Present
Bachelor of Science (Honours) in Computer Science *Malaysia*
Expected Graduation: October 2026 | CGPA: 3.83

Royal College Curepipe Jan 2018 – Nov 2022
Cambridge Assessment International Education *Mauritius*
A Levels & O Levels – Science Stream

Experience

Research Assistant – Server Development Apr 2026 – Present
HUMAC Lab, Sunway University — APNIC ISIF Asia 2025 *Malaysia*

- Engineered a backend platform for a LoRa-based hiker tracking system consisting of 20+ hikers and 4 LoRa towers, enabling real-time location monitoring in cellular-denied environments.
- Developed a LoRa data ingestion pipeline with MQTT and Node-RED to process tracker transmissions into structured GPS, device, battery, and SOS telemetry.
- Built geospatial services using NestJS, PostgreSQL/PostGIS, and GeoJSON for geofencing.
- Containerised backend infrastructure with Docker and designed RESTful and WebSocket APIs for communication between the mobile application, tracking devices, and server services.

Backend Developer Intern Jan 2026 – Apr 2026
Studio Twenty Sdn Bhd *Malaysia*

- Engineered 20+ RESTful API endpoints and 10+ PostgreSQL tables, enabling 5 core application features, by architecting modular services with NestJS and implementing structured data-access layers.
- Validated an 85% test-case pass rate in staging, improving pre-production reliability, by conducting API testing, debugging integration failures, and verifying end-to-end request/response workflows.
- Coordinated development within a 6-member engineering team, ensuring cross-platform feature integration across Android and iOS and alignment with project requirements, through API contract coordination, technical discussions, testing, and close collaboration with the project manager.

Values for Life Intern Mar 2023 – Apr 2023
Advisory Technology, PwC Moka *Mauritius*

- Developed and deployed 3+ features and bug fixes for an internal web application used by accounting and finance staff, by contributing to an existing ASP.NET, C#, and MySQL system through production-ready development and debugging.
- Improved user experience and interface usability, by redesigning navigation and page layouts to make workflows and information easier for internal users to understand and access.
- Resolved backend and database issues to improve application reliability and performance, by debugging server-side functionality, integrating MySQL data operations, and conducting code reviews with the project manager.

Projects

KAN-XGBoost Model Sentiment Analysis | *PyTorch, XGBoost, PyKAN* Sept. 2025 – Aug 2026

- Built a hybrid sentiment classifier achieving 87.5% accuracy by fusing DeBERTa-v3 semantic embeddings with an engineered-feature XGBoost structural pathway.
- Designed an interpretable Concept-Bottleneck KAN that decomposes predictions across 49 named sentiment concepts for improved model transparency.
- Engineered a leak-free adaptive fusion gate with calibrated outputs and a selective-prediction mode achieving 93.5% accuracy at 80% coverage.

Technical Skills

Languages: Java, Python, JavaScript, TypeScript
Backend Development: React, Node.js, Spring Boot, NestJS, PayloadCMS, ASP.NET
Developer Tools: Git, Docker, CI/CD pipelines
Testing: JUnit, Jest
AI/ML: Deep Learning Fundamentals, NLP